INCH-POUND

MIL-M-14/5
AMENDMENT 2
28 May 1991
SUPERSEDING
AMENDMENT 1
7 November 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, DIALLYL ISO-PHTHALATE THERMOSETTING, CONTAINING MINERAL FILLERS, GLASS OR POLYMERIC FILLERS

This amendment forms a part of MIL-M-14/5, dated 2 April 1990, and is approved for use by all Department and Agencies of the Department of the Defense.

PAGE 2

* Table I: Delete and substitute:

TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
			Mechanical/physical			
Compressive strength, endwise	24,000	24,000	20,000	20,000	24,000	24,000
Dimensional stability	0.1	0.1	0.2	0.2	0.2	0.2
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Heat deflection temperature	230	230	200	200	230	230
Heat resistance	160	160	120	120	140	140
Impact strength,						
Side[1]	2.75	2.75	0.35	0.35	0.50	0.50
Face[2]	3.00	3.00	0.35	0.35	0.50	0.50
Tensile strength	6,000	6,000	6,000	6,000	6,000	6,000
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5

See footnote at end of table.

AMSC N/A FSC 9330

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* Table I: Delete and substitute (page 3 attached):

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* Table II: Delete and substitute (page 4 attached):

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
Electrical						
Arc resistance	115	115	115	115	115	115
Dielectric breakdown:						
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	45	45	45	45	45	45
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	40	40	45	45	40	40
Dielectric constant:						
at 1 kilohertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.2	5.2	4.7	4.7
at 1 megahertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.2	5.2	4.7	4.7
Dielectric strength:						
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Dissipation factor:						
at 1 kilohertz	0.015	0.015	0.019	0.019	0.010	0.010
	0.018	0.018	0.020	0.020	0.013	0.013
at 1 megahertz	0.020	0.020	0.025	0.025	0.020	0.020
	0.023	0.023	0.025	0.025	0.023	0.023
Surface resistance	5,000	5,000	5.0	5.0	10	5,000
Volume resistance	5,000	5,000	2.0	2.0	10	5,000
Water extract conductors	60×10^{-6}	60×10^{-6}	–	–	–	–
Combustion						
Flame resistance:						
Ignition time	–	90	–	90	–	90
Burning time	–	90	–	90	–	90
Flammability/Thickness-Inch	–	V-0 / 1/8	–	V-0 / 1/16	–	V-0 / 1/16
Toxicity when heated:						
Carbon dioxide		15,000		15,000		
Carbon monoxide		1,000		1,000		
Ammonia		2,500		2,500		
Aldehydes as HCHO		50		50		
Cyanide as HCN		60		60		
Oxide of nitrogen as NO_2		100		100		
Hydrogen chloride		100		100		

[1]The side of the test specimen is that area formed by the chase of the mold.

[2]The face of a test specimen is that area formed by the top or bottom force plug.

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TABLE II. *Batch acceptance test requirements for diallyl iso-phthalate molding compounds, mineral/glass filled.*

Property to be tested	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
	Performance requirement for each compound.					
Arc resistance	115	115	115	115	115	115
Dielectric constant at 1 megahertz	4.7	4.7	5.2	5.2	4.7	4.7
Dielectric strength, step-by-step	300	300	300	300	300	300
Dissipation factor at 1 megahertz	0.023	0.023	0.025	0.025	.023	0.023
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Impact strength, side[1]	2.75	2.75	0.35	0.35	0.50	0.50
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5
Water extract conductors	60 x 10^{-6}	60 x 10^{-6}	–	–	–	–

[1]The side of the test specimen is that area formed by the chase of the mold.

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